

Credit Card Default Prediction

Reproducing and Improving a Supervised Machine Learning Study on Credit Card Default

Simon · Gabriel · Eduardo · Maria

DS 301 — Data Science Final Project

1. Introduction and Project Motivation

Credit card issuers need reliable ways to estimate whether a client will default on their next payment. This project addresses that problem as a binary classification task, using the public Default of Credit Card Clients dataset from the UCI Machine Learning Repository. The project was developed for the DS 301 Final Project and follows a published classification research paper as its methodological basis.

The project pursues two goals, as required by the assignment:

- Reproduce the methodology and main results presented in the research article.
- Improve on the original methodology while keeping the same dataset and the same three model families.

2. Research Paper Overview

"Predicting Default of Credit Card Clients Using Three Supervised Machine Learning Algorithms," by Thomson Ly, Rene Schuller, Katarina Simanic, and Sukhpreet Singh (University of Windsor), is the article this project reproduces and builds on.

The article investigates two research questions:

- Which hyperparameters have the strongest impact on each model?
- Which supervised machine learning model is the most effective at predicting default?

The article compares Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM), tuned with GridSearchCV. It concludes that SVM produced the best result among the three models, although the F1-score for the default class remained relatively low — a limitation this project revisits in its own contribution.

3. Dataset

The project uses the Default of Credit Card Clients dataset from the UCI Machine Learning Repository, containing payment, billing, credit, and demographic data for credit card clients in Taiwan.

Item	Value
Original observations	30,000
Removed observations	399
Final observations	29,601
Input features	23
Default rate	22.3%

Item	Value
Target	Default next month: 0 or 1

The 399 removed observations contained undocumented values in the EDUCATION or MARRIAGE variables, following the cleaning process described in the article.

4. Methodology

4.1 Article Reproduction

The reproduction follows the main steps described in the paper:

- Remove the ID column and undocumented categorical values.
- Split the data into 80% training and 20% testing.
- Standardize the features using StandardScaler.
- Apply random oversampling to the training data.
- Use 3-fold GridSearchCV to tune each model.
- Evaluate the models using accuracy, precision, recall, F1-score, and confusion matrices.

The complete article reproduction requires 6,156 model fits: 5,796 for Decision Tree, 342 for KNN, and 18 for SVM.

4.2 Group Contribution

Our contribution improves the evaluation process without changing the dataset or the three model families, addressing the assignment's requirement to experiment with different model parameters and create an upgraded version of the existing methodology:

- One-Hot Encoding — SEX, EDUCATION, and MARRIAGE are treated as categorical features instead of continuous numerical values, as suggested for future work in the article.
- Fold-Safe Oversampling — oversampling happens inside each cross-validation training fold, reducing the risk of data leakage between training and validation data.
- F1-Focused Model Selection — hyperparameters are selected using the default-class F1-score instead of accuracy, since the dataset is imbalanced and the default class is hardest to predict.
- Additional Hyperparameters — the improved experiments also evaluate minimum samples per leaf, class weights, additional K values, and SVM gamma settings.

5. Results

5.1 Default-Class F1-Score: Article vs. Reproduction vs. Improved Method

Model	Article	Our Reproduction	Improved Method
Decision Tree	0.410	0.392	0.523
KNN	0.390	0.380	0.469
SVM	0.520	0.514	0.534

Our reproduction follows the same general pattern as the article: SVM is the strongest model. The small differences are expected because the article does not provide its random seed or every implementation detail.

5.2 Improved Method — Full Metrics

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.768	0.483	0.572	0.523
KNN	0.681	0.373	0.632	0.469
SVM	0.776	0.498	0.575	0.534

5.3 Main Result Highlights

- SVM remained the strongest overall model.
- The reproduced SVM obtained accuracy of approximately 0.760 and F1-score of 0.514.
- The improved SVM increased the F1-score to 0.534.
- Decision Tree showed the largest F1 improvement, increasing from 0.392 to 0.523.
- KNN achieved the highest recall, 0.632, but produced lower precision and accuracy.
- The improved method increased the default-class F1-score for all three models.

These results show that the best model depends on the project objective: KNN detects more default clients, while SVM provides a better overall balance between precision, recall, F1-score, and accuracy. This directly answers the assignment's requirement to argue whether the group's contribution improved model performance: it did, consistently, across all three model families.

6. Challenges

6.1 Class Imbalance

Only around 22.3% of clients are in the default class. Because of this imbalance, a model can show acceptable accuracy while still missing many clients who will default.

6.2 Exact Reproduction

The article does not provide its random seed or every software and implementation detail. Our results are therefore close to the published results but not exactly identical.

6.3 Computational Cost

The original hyperparameter grids require thousands of model fits. SVM is especially expensive because it processes a large dataset with different kernels and values of C.

6.4 Data Leakage Risk

Applying oversampling before cross-validation can allow repeated observations to appear in different folds. The improved pipeline addresses this by applying oversampling only inside each training fold.

6.5 Metric Trade-Offs

Improving recall can create more false positives and reduce precision or accuracy. In a real financial project, the cost of each error would need to be considered before selecting a final model.

7. Main Learnings

- Accuracy is not enough for an imbalanced classification problem.
- Precision, recall, and F1-score provide important information about the minority class.
- Preprocessing must be fitted only on training data to avoid data leakage.
- Oversampling should be included inside the cross-validation pipeline.
- Categorical variables should be encoded according to their real meaning.
- GridSearchCV helps compare hyperparameters systematically, but it can require significant computational time.
- The metric used during model selection should match the real objective of the project.
- Reproducing a scientific article does not always produce identical numbers; a clear and documented methodology is essential for explaining the differences.

8. Limitations and Future Work

- The dataset represents clients from one Taiwanese bank and one historical period.
- The article does not provide all details needed for an exact reproduction.
- Random oversampling can increase overfitting because it repeats minority-class observations.
- The project does not include probability calibration or financial cost analysis.
- The final F1-score is still moderate, so the model should not be used alone for automatic credit decisions.

Future work could compare standard scaling with min-max scaling, test 5-fold cross-validation, evaluate SMOTE and other imbalance-handling methods, tune classification thresholds, add probability calibration, compare Random Forest, XGBoost, and neural networks, build a financial cost matrix for false positives and false negatives, and test the methodology on a more recent credit-risk dataset.

9. Conclusion

This project reproduced the Decision Tree, KNN, and SVM methodology of Ly, Schuller, Simanic, and Singh (University of Windsor) on the UCI Default of Credit Card Clients dataset, confirming SVM as the strongest of the three models. Our group contribution — one-hot encoding, fold-safe oversampling, and F1-focused hyperparameter selection — improved the default-class F1-score for every model while keeping the same dataset and model families, satisfying the assignment's requirement for a significant, well-argued contribution. The improved SVM reached an F1-score of 0.534, up from 0.514 in our reproduction and 0.520 in the original article. The score remains moderate, so the model should be treated as a decision-support tool rather than an automatic credit-approval system.

References

1. Ly, T., Schuller, R., Simanic, K., & Singh, S. Predicting Default of Credit Card Clients Using Three Supervised Machine Learning Algorithms. University of Windsor.

2. UCI Machine Learning Repository. Default of Credit Card Clients.
<https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>
3. Yeh, I.-C., & Lien, C.-H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473-2480.
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